

Hierarchical symmetry selects log-Poisson cascades: classification, uniqueness, and stability

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Abstract

Within i.i.d. multiplicative cascades, a single axiom—the hierarchical symmetry, a linear contraction on incremental scaling exponents—is shown to be necessary and sufficient for the cascade multiplier to be log-Poisson. We prove: (1) a characterization theorem determining the log-Poisson law with explicit parameters, within the class of *all* multipliers with finite lattice moments; (2) a classification theorem locating the log-Poisson class inside the log-infinitely-divisible family and identifying how every rival sub-family fails the symmetry; (3) a stability theorem with explicit constants— $(1 + \beta)^{1/2}$ known increment, $\sqrt{2}$ fitted, sharp at the variance level—and (4) an unconditional propagation theorem transferring the bound to the multiplier distribution at the sharp rate $\Theta(\sqrt{\varepsilon})$, with a matching lower bound. Beyond independence, the classification extends *exactly* at the level of asymptotic statistics (the limiting cumulant generating function at lattice orders, with large deviations and the multifractal spectrum under off-lattice regularity) and *provably not* at the level of laws: an explicit stationary ergodic Markov multiplier satisfies the symmetry exactly with a non-log-Poisson marginal, while exchangeable multipliers with exact scaling collapse to the i.i.d. log-Poisson cascade and finite-state Markov multipliers cannot satisfy the symmetry’s real-argument form with nontrivial intermittency. For exactly scale-invariant log-infinitely-divisible multifractal random measures, five exact contiguous moment orders identify the cascade class while four contiguous orders admit impersonators, and at the level of the scale-invariance generator the symmetry selects exactly the Barral–Mandelbrot compound Poisson cascade, with scale-ratio-free stability constants. The proofs reduce to second-moment identities and multiplicative couplings.

Keywords: Multiplicative cascades, Log-Poisson distribution, Hierarchical symmetry, Multifractal spectrum, Hausdorff moment problem, Multifractal random measures, Large deviations

1 Introduction

Multiplicative cascades model the successive fragmentation of a conserved quantity across scales and arise in fully developed turbulence [13, 14], rainfall, finance, and other settings exhibiting intermittent, scale-invariant fluctuations. The mathematical foundations of multiplicative cascades were established by Kahane and Peyrière [10]; see Barral and Mandelbrot [4] and Bacry and Muzy [2] for the modern discrete and continuous theories. The statistical properties of a cascade are encoded in the scaling exponents ζ_p of the structure functions $S_p(\ell) = \langle |\Phi(\ell)|^p \rangle \sim \ell^{\zeta_p}$. A central question is: *which probability distributions on the cascade multiplier W are compatible with observed scaling laws?*

Kolmogorov [13] proposed log-normal multipliers, leading to quadratic scaling exponents. Z.-S. She and Lévéque [16] introduced a hierarchical symmetry for the scaling exponents and derived a different, log-Poisson exponent formula that has since shown excellent agreement with experimental data. Dubrulle [7] independently identified the log-Poisson form. Z.-S. She and Waymire [17] used the Lévy–Khintchine representation to argue that this symmetry selects log-Poisson within the log-infinitely-divisible family; Dubrulle and Graner [8] reached a similar conclusion via symmetry groups. These works provided compelling physical arguments but did not supply rigorous proofs. Z.-S. She and Zhang [18] subsequently proposed that the hierarchical symmetry is *universal*—applicable not only to turbulence but to general multi-scale fluctuation systems including MHD turbulence, natural image statistics, and biological signals—and should serve as a standard analytical framework. The present paper supplies the rigorous mathematical foundation for this program.

We formalize the hierarchical symmetry as a single axiom (A1) and prove the following.

Characterization (Theorem 3). A1 uniquely determines the cascade multiplier W to be log-Poisson, with parameters ($a = \gamma \ln r$, $b = (\ln \beta)/k$, $\lambda = -C \ln r$) expressed in terms of the observable scaling exponents. No other distribution—infinately divisible or not—is compatible with A1.

Classification (Theorem 7). Within the full log-infinitely-divisible family, A1 selects exactly the log-Poisson class, and the proof stratifies the exclusion: Gaussian components, positive jumps, and stable generators of index $\alpha \in [1, 2)$ are eliminated by divergence of the incremental exponents; all remaining generators are eliminated by a second-moment rigidity.

Stability (Theorem 10). If A1 holds only approximately, with residuals bounded by ε , then the (tilted, compactified) Lévy measure of the generator is within $K\sqrt{\varepsilon}$ of a Dirac mass in the Wasserstein-1 metric, with the constant $K = ((1 + \beta)|\ln r|/|A|)^{1/2}$ when the limiting increment δ_∞ is known, and the constant $(2|\ln r|/|A|)^{1/2}$ when δ_∞ is fitted; both are sharp for the underlying variance bounds.

Propagation at the sharp rate (Theorems 11 and 12). Closeness of the Lévy measure transfers to closeness of the multiplier distribution at rate $O(\sqrt{\varepsilon})$ *unconditionally*—no finite-activity or minimum-jump hypothesis—with an explicit constant; and an explicit family shows the rate $\Theta(\sqrt{\varepsilon})$ is exact.

Beyond independence (Section 6). For stationary ergodic multipliers, A1 is equivalent to the limiting cumulant generating function—hence the large-deviation rate and the multifractal spectrum—being exactly log-Poisson (Theorem 13); but it provably does not determine the multiplier law: an explicit stationary ergodic countable-state Markov multiplier satisfies A1 exactly with a non-log-Poisson marginal (Theorem 14). Exchangeable multipliers with exact scaling collapse to the i.i.d. log-Poisson cascade (Corollary 15), and finite-state Markov multipliers cannot satisfy the real- p form of A1 with nontrivial intermittency (Theorem 16).

Continuous cascades (Section 7). In the Bacry–Muzy category of exactly scale-invariant log-infinitely-divisible multifractal random measures, identifiability of the cascade class from structure-function exponents has a sharp finite-order boundary: *four contiguous orders admit impersonators, five exact contiguous orders force the compound Poisson cascade* (Theorem 17); at the level of the scale-invariance generator—magnitude statistics, observable at all orders—A1 selects exactly the Barral–Mandelbrot compound Poisson cascade (Theorem 18), with stability constants that are native and scale-ratio-free (Corollary 19).

The converse—that log-Poisson multipliers imply A1—is established in Proposition 5, yielding a biconditional equivalence (Corollary 6).

Relation to prior work. The exponent formula **(**)** (Lemma 1) and the log-Poisson identification are due to Z.-S. She and Lévy [16] and Dubrulle [7]. Z.-S. She and Waymire [17] gave the first argument connecting A1 to the Lévy–Khintchine classification. The compound Poisson cascades appear in Barral–Mandelbrot [4]; the log-ID multifractal random measures in Bacry–Muzy [2, 15]; magnitude-cumulant analysis in Delour–Muzy–Arneodo [5]. The following results are new: the boundedness and moment-determinacy lemma (Lemma 2); the converse and biconditional (Proposition 5, Corollary 6); the classification with stratified exclusion (Theorem 7); the determinacy dichotomy (Proposition 9); the stability theorem with sharp constants under both readings (Theorem 10); the unconditional sharp-rate propagation theory (Theorems 11, 12); the beyond-i.i.d. classification and its impossibility boundary (Theorems 13, 14, 16, Corollary 15); and the continuous-cascade results (Theorems 17, 18).

Method. The change of variables $u = e^{kx}$ maps the Lévy measure from $(-\infty, 0]$ to the compact interval $[0, 1]$, where a second-moment identity against the candidate atom decides the classification and its stability. A1 forces the multiplier to be essentially bounded with $\text{ess sup } W = r^\gamma$, which yields moment determinacy directly and eliminates all tail estimates from the propagation argument; the propagation itself is a multiplicative coupling in which jumps near the identity are costed by their multiplicative deviation rather than counted. Beyond independence the arguments run at the level of limiting cumulant generating functions; in the continuous category they transfer through the dictionary $\ln r \mapsto -1$.

Throughout this paper, $\log$ and $\ln$ both denote the natural logarithm.

2 Setup

Let $r \in (0, 1)$ be a scale ratio. A *multiplicative cascade* generates a random positive measure μ on nested sets $B_0 \supset B_1 \supset \dots$ via

$$\mu(B_{n+1}) = W_{n+1} \cdot \mu(B_n),$$

where $\{W_n\}$ are i.i.d. positive random variables with $\mathbb{E}[W] = 1$ (conservation of mean; a normalization convention—none of the proofs below depend on it, cf. Corollary 20). Sections 6 and 7 relax, respectively, the independence assumption and the discrete setting.

Let $\Phi(\ell)$ be the cascade observable at scale $\ell = r^n$, operatively $\Phi(r^n) \stackrel{d}{=} W_1 \cdots W_n$ with $\Phi(1) = 1$. Define the *structure functions*

$$S_p(\ell) = \langle |\Phi(\ell)|^p \rangle = \ell^{\zeta_p},$$

where ζ_p are the *scaling exponents*, with $\zeta_0 = 0$; all moments of W are assumed finite, so that ζ_p is finite for every $p \geq 0$. The relation $\mathbb{E}[W^p] = r^{\zeta_p}$ identifies the single-step moment structure of the multiplier with the observable's scaling.

For a fixed integer $k \geq 1$ (the *hierarchy step*), define the *moment ratios*

$$H_p(\ell) = \frac{S_{p+k}(\ell)}{S_p(\ell)} = \ell^{\delta_p},$$

where $\delta_p = \zeta_{p+k} - \zeta_p$ are the *incremental exponents* at step k .

3 The axiom

Axiom (A1: Hierarchical Symmetry) There exist $\beta \in (0, 1)$ and $L \in \mathbb{R}$ such that for all $p \in k\mathbb{N}_0$ (non-negative integer multiples of k), the incremental exponents satisfy

$$\delta_{p+k} = (1 - \beta)L + \beta\delta_p. \quad (*)$$

Since $(*)$ is a contraction, it forces $\delta_{mk} \rightarrow L$ as $m \rightarrow \infty$; we write $\delta_\infty := L$. (Stating the axiom with a free constant L , rather than defining δ_∞ as a limit inside the equation that uses it, matters only for the approximate version in Theorem 10, where the distinction between the true limit and a fitted constant is quantitatively significant.)

A1 determines the full parameter set from the observable exponents:

Parameter	Determined by	Meaning
β	Contraction ratio of $(*)$	Coupling strength
γ	δ_∞/k	Linear drift
C	$(\delta_0 - \delta_\infty)/(1 - \beta)$	Concentration amplitude

Edge case ($C = 0$). If $\delta_0 = \delta_\infty$, then $C = 0$ and $\zeta_p = \gamma p$ (monofractal scaling). The cascade multiplier $W = r^\gamma$ is deterministic. A1 is trivially satisfied for any $\beta \in (0, 1)$. The classification and stability theorems assume $C > 0$ (nontrivial intermittency).

4 Results: the i.i.d. cascade

Lemma 1 (Exponent Form) *If the incremental exponents $\{\delta_p\}$ satisfy the A1 recurrence $(*)$ with $\beta \in (0, 1)$, then with $\zeta_0 = 0$:*

$$\zeta_p = \gamma p + C(1 - \beta^{p/k}), \quad (**)$$

where $\gamma = \delta_\infty/k$ and $C = (\delta_0 - \delta_\infty)/(1 - \beta)$.

We emphasize that this lemma is purely algebraic and involves no probabilistic content.

Proof (1) The recurrence $(*)$ is first-order linear with fixed point δ_∞ :

$$\delta_{p+k} - \delta_\infty = \beta(\delta_p - \delta_\infty).$$

(2) At $p = mk$ (integer multiples of k), iteration gives

$$\delta_{mk} - \delta_\infty = (\delta_0 - \delta_\infty) \beta^m.$$

Replacing $m = p/k$:

$$\delta_p = \delta_\infty + (\delta_0 - \delta_\infty) \beta^{p/k}.$$

(This formula is derived at $p \in k\mathbb{N}_0$. For general $p \geq 0$, we define ζ_p by $(**)$; the function $\beta^{p/k}$ is well-defined for all real $p \geq 0$ since $\beta > 0$. The moment-based arguments in Lemma 2 and Theorem 3 use only the lattice $p \in k\mathbb{N}_0$, where the formula is proved.)

(3) With $\zeta_0 = 0$, sum the step- k increments using the geometric series:

$$\zeta_p = \frac{p}{k} \delta_\infty + \frac{\delta_0 - \delta_\infty}{1 - \beta} (1 - \beta^{p/k}).$$

Identifying $\gamma = \delta_\infty/k$ and $C = (\delta_0 - \delta_\infty)/(1 - \beta)$:

$$\zeta_p = \gamma p + C(1 - \beta^{p/k}). \quad \square$$

Lemma 2 (Boundedness and Moment Determinacy) *Let $\{W_n\}$ be an i.i.d. multiplicative cascade whose lattice scaling exponents satisfy $\zeta_{km} = \gamma km + C(1 - \beta^m)$ for all $m \in \mathbb{N}_0$, as produced by A1 via Lemma 1. Then:*

- (i) W is essentially bounded, with $\text{ess sup } W = r^\gamma$;
- (ii) the law of W is uniquely determined by the lattice moments $\{\mathbb{E}[W^{km}]\}_{m \geq 0}$.

Proof Set $V = W^k \geq 0$. By independence across cascade levels, $\mathbb{E}[(W_1 \cdots W_n)^p] = (\mathbb{E}[W^p])^n$; combined with $S_p(\ell) = r^{n\zeta_p}$ this gives, for a single step, $\mathbb{E}[V^m] = \mathbb{E}[W^{km}] = r^{\zeta_{km}}$ for every $m \in \mathbb{N}_0$ —these are exactly the moments constrained by A1. Then

$$(\mathbb{E}[V^m])^{1/m} = r^{\zeta_{km}/m} = r^{\gamma k + C(1 - \beta^m)/m} \longrightarrow r^{\gamma k} \quad (m \rightarrow \infty).$$

On a probability space the norms $\|V\|_{L^m}$ are nondecreasing in m and converge to $\|V\|_{L^\infty}$; hence V is essentially bounded with $\|V\|_\infty = r^{\gamma k}$, and $W = V^{1/k}$ is bounded with $\|W\|_\infty = r^\gamma$, proving (i).

For (ii): a probability law supported in the compact interval $[0, r^{\gamma k}]$ is uniquely determined by its integer moments. (Hausdorff moment problem: polynomials are uniformly dense in $C([0, r^{\gamma k}])$ by the Weierstrass approximation theorem, so two laws with equal moments integrate every continuous function equally and coincide by the Riesz representation theorem.) Hence the law of V is determined by $\{\mathbb{E}[V^m]\}$; since $x \mapsto x^{1/k}$ is a Borel bijection of $[0, \infty)$, the law of W is determined as well. $\square$

Remark A Carleman-condition argument [1] is available here only along the lattice (applied to $V = W^k$), since A1 constrains no other moments when $k \geq 2$. The boundedness route above is shorter and stronger: statement (i) identifies the essential supremum of the multiplier with the most-singular scaling factor r^γ , a fact used again in Theorem 11.

Theorem 3 (Characterization) *Let $\{W_n\}$ be an i.i.d. multiplicative cascade whose incremental scaling exponents satisfy A1. Then:*

(i) **Scaling exponents.**

$$\zeta_p = \gamma p + C(1 - \beta^{p/k}),$$

where $\gamma = \delta_\infty/k$ and $C = (\delta_0 - \delta_\infty)/(1 - \beta)$.

(ii) **Uniqueness.** *The cascade multiplier W is uniquely determined to be log-Poisson:*

$$\log W = a + bN, \quad N \sim \text{Poisson}(\lambda),$$

$$a = \gamma \ln r, \quad b = \frac{\ln \beta}{k}, \quad \lambda = -C \ln r.$$

No other probability distribution on W is compatible with A1.

(iii) **Multifractal spectrum.** *Let d denote the spatial dimension of the cascade support. Then*

$$f(h) = d - C + Cx(1 - \ln x), \quad x = \frac{k(h - \gamma)}{C|\ln \beta|},$$

defined for $h \in [\gamma, \gamma + (C/k)|\ln \beta|]$.

Proof (i) Immediate from Lemma 1.

(ii) Set $a = \gamma \ln r$, $b = (\ln \beta)/k$, $\lambda = -C \ln r \geq 0$; here $C \geq 0$ is forced, since the moment sequence $m \mapsto \mathbb{E}[W^{km}]$ is log-convex and by (**) its second logarithmic difference is $C(1 - \beta)^2 \beta^m |\ln r|$ ($C = 0$ is the degenerate case $N \equiv 0$). For $\log W = a + bN$ with $N \sim \text{Poisson}(\lambda)$:

$$\mathbb{E}[W^p] = e^{ap} \cdot \exp[\lambda(e^{bp} - 1)] \quad \text{for all real } p \geq 0,$$

and since $e^{bp} = \beta^{p/k}$, the choice $\lambda = -C \ln r$ gives $\lambda(\beta^{p/k} - 1) = C \ln r (1 - \beta^{p/k})$, so that

$$\mathbb{E}[W^{km}] = e^{(\gamma \ln r)km} e^{C \ln r (1 - \beta^m)} = r^{\zeta_{km}} \quad \text{for every } m \in \mathbb{N}_0.$$

Thus the log-Poisson law realizes exactly the lattice moments of the cascade multiplier. By Lemma 2(ii) the lattice moments uniquely determine the law; hence W is log-Poisson, and no other distribution is possible. (The same matching holds at every real $p \geq 0$, so the extension of (**) off the lattice is consistent.)

(iii) The singularity spectrum $f(h) = \inf_p [ph - \zeta_p + d]$. Setting the derivative to zero:

$$h - \gamma + \frac{C}{k}(\ln \beta) \beta^{p/k} = 0 \implies h - \gamma = \frac{C|\ln \beta|}{k} \beta^{p/k}.$$

Define $x = \beta^{p/k} = k(h - \gamma)/(C|\ln \beta|)$. Then $p = -k \ln x / |\ln \beta|$ and

$$p(h - \gamma) = \frac{-k \ln x}{|\ln \beta|} \cdot \frac{C|\ln \beta|}{k} x = -Cx \ln x.$$

Therefore

$$f(h) = d - C + Cx(1 - \ln x), \quad x = \frac{k(h - \gamma)}{C|\ln \beta|}.$$

Boundary checks: $p = 0 \Rightarrow x = 1 \Rightarrow f = d$; $p \rightarrow \infty \Rightarrow x \rightarrow 0 \Rightarrow f \rightarrow d - C$. Concavity: $\zeta_p'' = -(C/k^2)(\ln \beta)^2 \beta^{p/k} < 0$. $\square$

Corollary 4 (Most-singular branch) *Under A1, $W \leq r^\gamma$ almost surely, and the bound is attained with positive probability: $\mathbb{P}(W = r^\gamma) = e^{-\lambda}$. The probability that a cascade trajectory takes the maximal factor for n consecutive levels is*

$$e^{-\lambda n} = r^{Cn} = \ell^C \quad \text{at scale } \ell = r^n :$$

the set of always-maximal cascade paths carries codimension exactly C , in agreement with $f(h_{\min}) = d - C$ in Theorem 3(iii).

Proof By Theorem 3(ii), $W = r^\gamma \beta^{N/k}$ with $N \sim \text{Poisson}(\lambda)$, so $W \leq r^\gamma$ with equality iff $N = 0$, an event of probability $e^{-\lambda}$. By independence across levels, n consecutive maximal factors have probability $e^{-\lambda n} = e^{(C \ln r)n} = r^{Cn}$. $\square$

Remark (Scope of Theorem 3) No infinite-divisibility assumption enters Theorem 3: A1 characterizes the log-Poisson law within the class of *all* nonnegative multipliers with finite lattice moments. Theorem 7 below is therefore not a larger uniqueness statement but an anatomical one: it locates the log-Poisson class inside the Lévy–Khintchine parameterization and identifies the mechanism by which each rival sub-family fails A1. Its proof technique—the compactifying substitution $u = e^{kx}$ —is also the engine of the stability and propagation theory.

Remark (Conservation) The setup assumes $\mathbb{E}[W] = 1$, which requires $\zeta_1 = 0$. Substituting into (**): $\gamma + C(1 - \beta^{1/k}) = 0$, giving $\gamma = -C(1 - \beta^{1/k})$. This is a constraint relating γ to C and β , reducing the free parameters from three to two.

Proposition 5 (Converse) *If the cascade multiplier W is log-Poisson—that is, $\log W = a + bN$ with $N \sim \text{Poisson}(\lambda)$, $b < 0$, $\lambda > 0$ —then the incremental scaling exponents satisfy A1 with $\beta = e^{bk} \in (0, 1)$.*

Proof The moment generating function gives $\mathbb{E}[W^p] = \exp(ap + \lambda(e^{bp} - 1))$, so $\zeta_p = (ap + \lambda(e^{bp} - 1))/\ln r$. The step- k increments are

$$\delta_p = \frac{ak + \lambda e^{bp}(e^{bk} - 1)}{\ln r}.$$

Setting $\beta = e^{bk} \in (0, 1)$ (since $b < 0$, $k \geq 1$):

$$\delta_p = \frac{ak}{\ln r} + \frac{\lambda(\beta - 1)}{\ln r} \beta^{p/k}.$$

As $p \rightarrow \infty$: $\beta^{p/k} \rightarrow 0$, so $\delta_\infty = ak/\ln r$. The deviation is

$$\delta_p - \delta_\infty = \frac{\lambda(\beta - 1)}{\ln r} \beta^{p/k}.$$

At $p + k$:

$$\delta_{p+k} - \delta_\infty = \frac{\lambda(\beta - 1)}{\ln r} \beta^{(p+k)/k} = \beta(\delta_p - \delta_\infty).$$

Therefore $\delta_{p+k} = (1 - \beta)\delta_\infty + \beta\delta_p$, which is exactly A1. $\square$

Corollary 6 (Biconditional) *Within i.i.d. multiplicative cascades, A1 is necessary and sufficient for log-Poisson:*

$$A1 \text{ holds} \iff W \text{ is log-Poisson (with } b < 0\text{)}.$$

The forward direction is Theorem 3(ii); the reverse is Proposition 5.

Theorem 7 (Log-ID Classification) *Let $\{W_n\}$ be an i.i.d. multiplicative cascade with non-trivial intermittency ($C > 0$), whose generator $\log W$ is infinitely divisible with Lévy triplet (a, σ^2, ν) . Then A1 holds with $\beta \in (0, 1)$ if and only if $\sigma^2 = 0$ and $\nu = \lambda\delta_b$ for some $b < 0$, $\lambda > 0$. That is:*

A1 selects exactly the log-Poisson class from the full log-infinitely-divisible family.

No other log-ID cascade—log-normal, log-stable, or any intermediate—satisfies A1.

Proof Reverse direction. If $\nu = \lambda\delta_b$ with $b < 0$ and $\sigma^2 = 0$, then $\log W = a + bN$ with $N \sim \text{Poisson}(\lambda)$, and A1 holds by Proposition 5.

Forward direction. Assume A1 holds. We show $\sigma^2 = 0$ and $\nu = \lambda\delta_b$.

Step 1 (unsplit form). The cumulant generating function of $\log W$ is

$$\psi(p) = ap + \frac{\sigma^2 p^2}{2} + \int (e^{px} - 1 - px \mathbf{1}_{|x| \leq 1}) \nu(dx),$$

finite for all $p \geq 0$ since all moments of W are finite. With $\zeta_p = \psi(p)/\ln r$ and $\delta_p = (\psi(p+k) - \psi(p))/\ln r$, define $\phi(p) = \psi(p+k) - \psi(p)$. Then

$$\phi(p) = ak + \sigma^2 k \left(p + \frac{k}{2}\right) + \int g_p(x) \nu(dx),$$

$$g_p(x) := e^{px}(e^{kx} - 1) - kx \mathbf{1}_{|x| \leq 1},$$

where g_p is ν -integrable for each p , being the difference of the two compensated Lévy–Khinchine integrands. *No splitting of the integral is performed at this stage.*

Step 1' (sign inventory). For every $p \geq 0$:

- on $(0, 1]$: $e^{px} \geq 1$ gives $g_p(x) \geq (e^{kx} - 1) - kx \geq 0$, and $g_p(x) \uparrow \infty$ pointwise as $p \rightarrow \infty$;
- on $(1, \infty)$: $g_p(x) = e^{px}(e^{kx} - 1) \geq 0$, increasing to $+\infty$ pointwise;
- on $[-1, 0)$: $|e^{px}(e^{kx} - 1)| \leq 1 - e^{kx} \leq k|x|$, so $0 \leq g_p(x) \leq k|x|$, with $g_p(x) \rightarrow k|x|$ pointwise as $p \rightarrow \infty$;
- on $(-\infty, -1)$: $-1 \leq g_p(x) \leq 0$, with $g_p(x) \rightarrow 0$ pointwise.

In particular $\int g_p d\nu \geq -\nu((-\infty, -1))$, uniformly in p .

Step 2 ($\sigma^2 = 0$). If $\sigma^2 > 0$ then, by Step 1',

$$\phi(p) \geq ak + \sigma^2 k \left(p + \frac{k}{2}\right) - \nu((-\infty, -1)) \rightarrow +\infty,$$

so $\delta_p = \phi(p)/\ln r \rightarrow -\infty$, contradicting the finite limit δ_∞ forced by A1. Hence $\sigma^2 = 0$. *This eliminates all log-normal and mixed Gaussian-jump generators.*

Step 3 ($\text{supp}(\nu) \subseteq (-\infty, 0]$). If ν has mass on $(0, \infty)$ then, since $g_p \geq 0$ there and $g_p \uparrow \infty$ pointwise, monotone convergence gives $\int_{(0, \infty)} g_p d\nu \rightarrow \infty$, while $\int_{(-\infty, 0)} g_p d\nu \geq -\nu((-\infty, -1))$; again $\phi(p) \rightarrow \infty$ and $\delta_p \rightarrow -\infty$, a contradiction. Therefore $\text{supp}(\nu) \subseteq (-\infty, 0]$. *This eliminates all generators with positive jumps.*

Step 3₂ (integrability near 0). We claim A1 forces $\int_{[-1, 0)} |x| \nu(dx) < \infty$. With $\sigma^2 = 0$ and $\text{supp } \nu \subseteq (-\infty, 0]$, apply Fatou's lemma on $[-1, 0)$ (integrand $g_p \geq 0$ by Step 1', pointwise limit $k|x|$) and dominated convergence on $(-\infty, -1)$ (bounded by 1, finite mass):

$$\liminf_{p \rightarrow \infty} \phi(p) \geq ak + k \int_{[-1, 0)} |x| \nu(dx) - \nu((-\infty, -1)).$$

If $\int_{[-1, 0)} |x| d\nu = \infty$ then $\phi(p) \rightarrow \infty$ and $\delta_p \rightarrow -\infty$, contradicting A1. Hence $\int_{|x| \leq 1} |x| d\nu < \infty$, the compensator integral $k \int x \mathbb{1}_{|x| \leq 1} d\nu$ is finite, and *only now* may the integral be split:

$$\phi(p) = c_0 + \int_{(-\infty, 0)} e^{px} (e^{kx} - 1) \nu(dx), \quad c_0 = ak - k \int x \mathbb{1}_{|x| \leq 1} \nu(dx).$$

The remaining integrand is dominated by $1 - e^{kx} \leq \min(1, k|x|) \in L^1(\nu)$ and tends to 0 pointwise, so by dominated convergence $\phi(p) \rightarrow c_0$ as $p \rightarrow \infty$; thus $\phi_\infty = c_0$ and $\delta_\infty = c_0/\ln r$.

Step 4 (ν is a single Dirac mass). A1 at $p = mk$ gives, by the closed form of Lemma 1, $\delta_{mk} - \delta_\infty = (\delta_0 - \delta_\infty)\beta^m$; multiplying by $\ln r$,

$$\int_{(-\infty, 0)} e^{mkx} (e^{kx} - 1) \nu(dx) = A\beta^m \quad \text{for all } m \geq 0, \quad (1)$$

where $A = (\delta_0 - \delta_\infty) \ln r < 0$ (nontrivial intermittency and $\ln r < 0$). Substitute $u = e^{kx}$, mapping $(-\infty, 0) \rightarrow (0, 1)$; let $\tilde{\nu}$ be the pushforward of ν and set

$$\eta := (1 - u) d\tilde{\nu} \geq 0, \quad \mu_m := \int_{(0, 1)} u^m d\eta.$$

Then (1) reads $\mu_m = |A| \beta^m$ for all $m \geq 0$; the case $m = 0$ shows η is a finite positive measure of total mass $|A|$. Only $m \in \{0, 1, 2\}$ are needed:

$$\int_{(0, 1)} (u - \beta)^2 d\eta = \mu_2 - 2\beta\mu_1 + \beta^2\mu_0 = |A|(\beta^2 - 2\beta^2 + \beta^2) = 0.$$

Since $(u - \beta)^2 > 0$ on $(0, 1) \setminus \{\beta\}$ and $\eta \geq 0$, we conclude $\eta((0, 1) \setminus \{\beta\}) = 0$ and $\eta(\{\beta\}) = |A|$. Because $u - 1 \neq 0$ on $(0, 1)$, the tilt is invertible:

$$\tilde{\nu} = \frac{|A|}{1 - \beta} \delta_\beta = \lambda \delta_\beta, \quad \lambda = \frac{|A|}{1 - \beta} > 0.$$

Therefore $\nu = \lambda \delta_b$ with $b = (\ln \beta)/k < 0$ and $\lambda > 0$. The generator $\log W$ is compound Poisson with deterministic jump size b and rate λ : this is the log-Poisson distribution. $\square$

Remark (Alternative identification; minimality) The conclusion of Step 4 can also be reached from the full moment sequence: a finite signed measure on a compact interval is determined by its moments (Weierstrass approximation and the Riesz representation theorem), and $A\delta_\beta$ realizes the moments (1). The second-moment argument given above is preferred because it (a) uses only $m \in \{0, 1, 2\}$ of (1), so that A1 restricted to $p \in \{0, k, 2k\}$, together with Steps 2–3 $\frac{1}{2}$, already pins the distribution; and (b) is exactly the computation that the stability theorem quantifies (see the Remark closing Section 5).

Remark (Which families die where) The exclusion mechanism stratifies. Gaussian components (Step 2), positive jumps (Step 3), and negative-support Lévy measures with $\int_{|x| \leq 1} |x| d\nu = \infty$ —in particular totally skewed stable generators of index $\alpha \in [1, 2)$ —all fail A1 by *divergence*: $\delta_\infty = -\infty$ (Step 3 $\frac{1}{2}$). All remaining log-ID generators have bounded incremental exponents but fail the *geometric rigidity* of Step 4: e.g. for a stable generator of index $\alpha < 1$ the moments μ_m decay like the power law $m^{\alpha-1}$, which cannot equal $|A|\beta^m$ for any $\beta \in (0, 1)$.

Corollary 8 (Principal cascade classes) *The log-ID cascade family is partitioned by A1:*

<i>Class</i>	<i>Lévy data</i>	<i>A1</i>	<i>Failure mode</i>	<i>Determinate</i>
<i>Log-Poisson</i>	$\sigma^2 = 0,$ $\nu = \lambda \delta_b, b < 0$	<i>Holds</i>	—	<i>Yes (bounded W)</i>
<i>Log-normal</i>	$\sigma^2 > 0$	<i>Fails</i>	$\delta_\infty = -\infty$ (Step 2)	<i>No</i>
<i>Log-stable, $\alpha \in [1, 2)$</i>	ν power-law	<i>Fails</i>	$\delta_\infty = -\infty$ (Step 3 $\frac{1}{2}$)	—
<i>Log-stable, $\alpha < 1$</i>	ν power-law	<i>Fails</i>	non-geometric decay (Step 4)	<i>Yes (bounded W)</i>
<i>General log-ID</i>	<i>any other</i>	<i>Fails</i>	Step 3 or Step 4	—

Determinacy in this family tracks boundedness of the multiplier, equivalently boundedness of $\{\delta_p\}$ (Lemma 2(i)): every negative-support generator with finite δ_∞ has compactly supported W , hence is moment-determinate. The operative dichotomy is bounded versus unbounded, with A1 strictly on the bounded side.

Proposition 9 (Determinacy Dichotomy) *The two principal cascade exponent laws are distinguished by moment determinacy:*

(a) *If A1 holds within a cascade (log-Poisson regime), then the scaling exponents uniquely determine the multiplier law (Theorem 3(ii)); indeed W is bounded and moment-determinate (Lemma 2).*

(b) *If the exponents are quadratic, $\zeta_p = c_1 p + c_2 p^2$ with $c_2 < 0$ ([13]/log-normal regime), then $\mathbb{E}[W^p] = \exp(\mu p + \sigma^2 p^2/2)$ with $\mu = c_1 \ln r$ and $\sigma^2 = 2c_2 \ln r > 0$, and this moment sequence is indeterminate: uncountably many distinct laws realize it. Under quadratic scaling, the lattice exponents cannot identify the multiplier law.*

Proof Part (a) is contained in Lemma 2 and Theorem 3(ii).

Part (b). We exhibit the family (Heyde [9]). Let f be the log-normal density with parameters (μ, σ^2) and, for $|c| \leq 1$, define

$$f_c(x) = f(x) \left[1 + c \sin\left(\frac{2\pi(\ln x - \mu)}{\sigma^2}\right) \right], \quad x > 0.$$

Substituting $\ln x = \mu + \sigma z$ with Z standard normal, for every $n \in \mathbb{N}_0$:

$$\begin{aligned} \int_0^\infty x^n f(x) \sin\left(\frac{2\pi(\ln x - \mu)}{\sigma^2}\right) dx &= e^{n\mu} \mathbb{E} \left[e^{n\sigma Z} \sin\left(\frac{2\pi Z}{\sigma}\right) \right] \\ &= e^{n\mu} e^{(n^2\sigma^2 - 4\pi^2/\sigma^2)/2} \sin(2\pi n) = 0, \end{aligned}$$

using $\mathbb{E}[e^{(\alpha+i\beta')Z}] = e^{(\alpha+i\beta')^2/2}$, whose imaginary part is $e^{(\alpha^2-\beta'^2)/2} \sin(\alpha\beta')$, with $\alpha = n\sigma$, $\beta' = 2\pi/\sigma$, $\alpha\beta' = 2\pi n$. The case $n = 0$ shows each f_c is a probability density (and $f_c \geq 0$ since $|c| \leq 1$); the cases $n \geq 1$ show all f_c share the log-normal moments $\exp(n\mu + n^2\sigma^2/2)$. Hence uncountably many distinct laws realize the moment sequence. $\square$

Remark Indeterminacy cannot be inferred from the convergence of the Carleman sum: Carleman's condition is sufficient for determinacy but not necessary, so its failure proves nothing. The explicit family above is the classical argument.

Theorem 10 (Stability) *Let $\{W_n\}$ be an i.i.d. multiplicative cascade with log-infinitely-divisible generator, all moments finite, and nontrivial intermittency. Suppose there exist $\beta \in (0, 1)$, $d^* \in \mathbb{R}$ and $\varepsilon > 0$ such that*

$$|\delta_{p+k} - (1 - \beta)d^* - \beta\delta_p| < \varepsilon \quad \text{for all } p \in k\mathbb{N}_0.$$

Then $\sigma^2 = 0$, $\text{supp } \nu \subseteq (-\infty, 0]$, $\int(|x| \wedge 1) d\nu < \infty$, the limit $\delta_\infty := \lim_{p \rightarrow \infty} \delta_p$ exists finitely, and $(1 - \beta)|d^ - \delta_\infty| \leq \varepsilon$. Moreover, with $u = e^{kx}$, $\eta = (1 - u) d\tilde{\nu}$, and $A = (\delta_0 - \delta_\infty) \ln r$:*

(i) *if $d^* = \delta_\infty$ (the true limit is known),*

$$W_1\left(\frac{\eta}{\|\eta\|}, \delta_\beta\right) \leq \left(\frac{(1 + \beta)|\ln r|}{|A|}\right)^{1/2} \sqrt{\varepsilon};$$

(ii) *in general (fitted d^*),*

$$W_1\left(\frac{\eta}{\|\eta\|}, \delta_\beta\right) \leq \left(\frac{2|\ln r|}{|A|}\right)^{1/2} \sqrt{\varepsilon}.$$

Both constants are sharp for the underlying variance bounds from which they arise (and attained in Wasserstein form for $\beta < 1/2$; see the remark below). In particular, the cascade multiplier distribution converges to log-Poisson as $\varepsilon \rightarrow 0$, at the sharp rate $\Theta(\sqrt{\varepsilon})$ (Theorems 11 and 12).

Proof Step 0 (reduction and existence of the limit). From the hypothesis, $|\delta_{(m+1)k}| \leq \beta|\delta_{mk}| + (1 - \beta)|d^*| + \varepsilon$, so the lattice sequence $\{\delta_{mk}\}$ is bounded: $\limsup_m |\delta_{mk}| \leq |d^*| + \varepsilon/(1 - \beta)$. But by Steps 2, 3 and $3\frac{1}{2}$ of the proof of Theorem 7—none of which used the exact form of A1, only the finiteness of $\liminf$ of $\{\delta_p\}$ —each of the conditions $\sigma^2 > 0$, $\nu((0, \infty)) > 0$, $\int_{|x| \leq 1} |x| d\nu = \infty$ forces $\delta_{mk} \rightarrow -\infty$, a contradiction. Hence $\sigma^2 = 0$, $\text{supp } \nu \subseteq (-\infty, 0]$, $\int(|x| \wedge 1) d\nu < \infty$; the split form of ϕ is valid, dominated convergence gives $\phi(p) \rightarrow c_0$, and $\delta_\infty = c_0/\ln r$ exists finitely. Letting $m \rightarrow \infty$ in the hypothesis,

$$|\delta_\infty - (1 - \beta)d^* - \beta\delta_\infty| \leq \varepsilon \implies (1 - \beta)|d^* - \delta_\infty| \leq \varepsilon.$$

Step 1 (exact moment identities and residuals). As in Step 4 of Theorem 7 (now with no approximation in the identity itself),

$$\mu_m := \int_{(0,1)} u^m d\eta = (\delta_{mk} - \delta_\infty) |\ln r| \geq 0 \quad \text{for all } m \geq 0,$$

in particular $\|\eta\| = \mu_0 = (\delta_0 - \delta_\infty) |\ln r| = |A|$ *exactly*. Define the signed residuals

$$\epsilon_m := \delta_{(m+1)k} - (1 - \beta) \delta_\infty - \beta \delta_{mk} = \frac{\mu_{m+1} - \beta \mu_m}{|\ln r|},$$

and let h_m denote the hypothesis residuals (with d^* in place of δ_∞), $|h_m| < \varepsilon$. Setting $t := (1 - \beta)(d^* - \delta_\infty)$, $|t| \leq \varepsilon$ by Step 0, one has $\epsilon_m = h_m + t$. Under reading (i), $t = 0$ and $|\epsilon_m| < \varepsilon$.

Step 2 (variance identity). Telescoping,

$$\begin{aligned} \int_{(0,1)} (u - \beta)^2 d\eta &= \mu_2 - 2\beta\mu_1 + \beta^2\mu_0 = (\mu_2 - \beta\mu_1) - \beta(\mu_1 - \beta\mu_0) \\ &= |\ln r| (\epsilon_1 - \beta\epsilon_0). \end{aligned}$$

Under reading (i): $|\epsilon_1 - \beta\epsilon_0| < (1 + \beta)\varepsilon$. Under reading (ii): $\epsilon_1 - \beta\epsilon_0 = (h_1 - \beta h_0) + (1 - \beta)t$, so $|\epsilon_1 - \beta\epsilon_0| < (1 + \beta)\varepsilon + (1 - \beta)\varepsilon = 2\varepsilon$. Hence

$$0 \leq \int_{(0,1)} (u - \beta)^2 d\eta \leq \begin{cases} (1 + \beta) |\ln r| \varepsilon & \text{(i),} \\ 2 |\ln r| \varepsilon & \text{(ii).} \end{cases}$$

Step 3 (Wasserstein bound). For a Dirac target the W_1 distance is the first absolute moment: $W_1(\eta/\|\eta\|, \delta_\beta) = \|\eta\|^{-1} \int |u - \beta| d\eta$. By Cauchy–Schwarz and $\|\eta\| = |A|$,

$$\begin{aligned} W_1\left(\frac{\eta}{\|\eta\|}, \delta_\beta\right) &\leq \left(\frac{1}{|A|} \int (u - \beta)^2 d\eta\right)^{1/2} \\ &\leq \begin{cases} ((1 + \beta) |\ln r| / |A|)^{1/2} \sqrt{\varepsilon} & \text{(i),} \\ (2 |\ln r| / |A|)^{1/2} \sqrt{\varepsilon} & \text{(ii).} \end{cases} \quad \square \end{aligned}$$

Remark (Sharpness; a warning about per-moment transfer) (a) The variance constant in reading (i) is attained in the limit by the two-atom family $\eta = c_0 \delta_{u_0} + c_1 \delta_v$ with $u_0 \downarrow 0$, $v \in (\beta, 1)$, $c_1 v(v - \beta) = \varepsilon |\ln r|$ and $c_0 = (\varepsilon |\ln r| / \beta)(1 + 1/v)$: then $\epsilon_0 \rightarrow -\varepsilon$, $\epsilon_1 \rightarrow +\varepsilon$, $|\epsilon_m| = \varepsilon v^{m-1} \leq \varepsilon$ for $m \geq 2$, and $\int (u - \beta)^2 d\eta \rightarrow (1 + \beta) |\ln r| \varepsilon$. The Wasserstein constant of the theorem follows by Cauchy–Schwarz; at $v = 2\beta$ (feasible for $\beta < 1/2$) the family is Cauchy–Schwarz-tight and attains it, while for $\beta \geq 1/2$ the Wasserstein form need not be attained.

(b) The variance constant in reading (ii) is likewise attained: choose moment residuals $(\epsilon_0, \epsilon_1) \rightarrow (0, 2\varepsilon)$ realized by two atoms (one near 0, one in $(\beta, 1)$) and the offset $t \rightarrow \varepsilon$; all hypothesis residuals $h_m = \epsilon_m - t$ then stay below ε in absolute value. At $v = 2\beta$ (again feasible for $\beta < 1/2$) this family is likewise Cauchy–Schwarz-tight and attains the Wasserstein constant of reading (ii). For $\beta < \sqrt{2} - 1$ one has $2 > (1 + \beta)^2$: the distinction between the two readings is quantitatively real, and a constant valid in reading (i)—even the non-sharp $(1 + \beta)^2$ —can *fail* outright in reading (ii).

(c) A tempting route passes through the per-moment estimate $|\mu_m - |A|\beta^m| \leq |\ln r| \varepsilon$ for all m . That estimate is false in general: recurrence errors accumulate to $|\mu_m - |A|\beta^m| \leq |\ln r| \varepsilon \frac{1 - \beta^m}{1 - \beta}$, and the bound is attained in the limit by $\eta = (|A| - c) \delta_\beta + c \delta_v$ with $c(v - \beta) = \varepsilon |\ln r|$, $v \rightarrow 1$, for which $\sup_m |\mu_m - |A|\beta^m| / (|\ln r| \varepsilon) \rightarrow 1/(1 - \beta)$. It is also unnecessary: only ϵ_0 and ϵ_1 enter the variance identity.

5 Propagation: the sharp rate

We now transfer the bound of Theorem 10 from the Lévy measure to the multiplier distribution, at a rate that is exact: $O(\sqrt{\varepsilon})$ with no further hypotheses (Theorem 11), and no better (Theorem 12).

Throughout this section the hypotheses are those of Theorem 10, reading (i) (for reading (ii) replace $(1 + \beta)$ by 2 in every constant), so that Step 0 there gives $\sigma^2 = 0$, $\text{supp } \nu \subseteq (-\infty, 0]$, $\int (|x| \wedge 1) d\nu < \infty$, and Steps 2–3 give, with $V_\varepsilon := (1 + \beta)|\ln r| \varepsilon$ and $S := ((1 + \beta)|A||\ln r|)^{1/2}$,

$$\int_{(0,1)} (u - \beta)^2 d\eta \leq V_\varepsilon, \quad \int_{(0,1)} |u - \beta| d\eta \leq S\sqrt{\varepsilon}. \quad (2)$$

The Lévy measure may have *infinite total mass*, accumulating only at $u = 1$ where the tilt $(1 - u)$ vanishes; the multiplier is the a.s.-convergent product over the Poisson point process $\{U_i\}$ of intensity $\tilde{\nu}$,

$$W = e^{a_\varepsilon} \prod_i U_i^{1/k}, \quad \sum_i (1 - U_i^{1/k}) \leq 2 \sum_i (1 - U_i) \text{ of finite mean } 2\|\eta\|$$

(Campbell's formula [12]). Under $\mathbb{E}[W] = 1$ the drift is $a_\varepsilon = \int (1 - u^{1/k}) d\tilde{\nu} < \infty$. The comparison target W_0 is the log-Poisson multiplier with jump factor $\beta^{1/k}$, rate $\lambda = |A|/(1 - \beta)$, and drift a_0 fixed by the same normalization.

Theorem 11 (Unconditional propagation) *Under the hypotheses of Theorem 10 alone—no finite-activity or minimum-jump assumption—there exist $\varepsilon_0 > 0$ and an explicit constant $K_\infty = K_\infty(\beta, C, r, k)$ such that for all $\varepsilon \leq \varepsilon_0$,*

$$W_1(\text{law}(W), \text{law}(W_0)) \leq K_\infty \sqrt{\varepsilon}.$$

One admissible (not optimized) choice is

$$K_\infty = 4e^{a_0+1} \left(\frac{L_k}{1-\beta} + \frac{1}{(1-\beta)^2} \right) \sqrt{(1+\beta)|A||\ln r|},$$

$$L_k = \frac{1}{k} \left(\frac{\beta}{2} \right)^{(1-k)/k},$$

with the $O(\varepsilon)$ remainders absorbed for $\varepsilon \leq \varepsilon_0$; at $k = 1$ the formalization verifies the sharper constant with e^{a_0} in place of e^{a_0+1} , under the explicit threshold $\varepsilon_0 = \|\eta\|/(4(1+\beta)|\ln r|)$ (the formalization's residual is measured at the Lévy-Khintchine scale $|\ln r| \varepsilon$).

Proof Fix the ε -independent split height

$$h_0 := \frac{1-\beta}{2}, \quad u_0 := 1 - h_0 = \frac{1+\beta}{2},$$

and call a jump *small* if $u \in (u_0, 1)$, *macroscopic* if $u \in (0, u_0]$.

Step 1 (small jumps: individually cheap, collectively $O(\varepsilon)$). Every $u \in (u_0, 1)$ lies at distance $> h_0$ from β , so by Chebyshev's inequality against (2),

$$\eta((u_0, 1)) \leq \frac{V_\varepsilon}{h_0^2} = \frac{4(1+\beta)|\ln r|}{(1-\beta)^2} \varepsilon.$$

Leave the small-jump points of W *unpaired* and cost them multiplicatively: for factors in $(0, 1]$, telescoping gives $|\prod_i s_i - 1| \leq \sum_i (1 - s_i)$ (valid for infinite products by monotone limits), and $1 - u^{1/k} \leq \frac{2}{k}(1 - u)$ for $u \geq \frac{1}{2}$ (derivative bound; $u_0 \geq \frac{1}{2}$). By Campbell's formula,

$$\begin{aligned} \mathbb{E} \left| \prod_{\text{small}} U_i^{1/k} - 1 \right| &\leq \frac{2}{k} \int_{(u_0, 1)} (1 - u) d\tilde{\nu} = \frac{2}{k} \eta((u_0, 1)) \\ &\leq \frac{8(1 + \beta) |\ln r|}{k(1 - \beta)^2} \varepsilon. \end{aligned}$$

This is the step that controls infinite activity: a near-1 jump's cost is its multiplicative deviation $\asymp (1 - u)$ —already η -weighted—not the count 1.

Step 2 (the macroscopic part is automatically finite-activity, with a fixed de-tilting constant). On $(0, u_0]$: $\frac{1}{1-u} \leq \frac{1}{h_0} = \frac{2}{1-\beta}$, so $\lambda_{\text{mac}} := \tilde{\nu}((0, u_0]) \leq \frac{2|A|}{1-\beta} < \infty$. Comparing rates against $\lambda = |A|/(1 - \beta) = \int \frac{d\eta}{1-\beta}$, and using $|\frac{1}{1-u} - \frac{1}{1-\beta}| = \frac{|u-\beta|}{(1-u)(1-\beta)} \leq \frac{2|u-\beta|}{(1-\beta)^2}$ on $(0, u_0]$,

$$|\lambda_{\text{mac}} - \lambda| \leq \frac{2}{(1-\beta)^2} S\sqrt{\varepsilon} + \frac{\eta((u_0, 1))}{1-\beta},$$

the second term being $O(\varepsilon)$ by Step 1.

Step 3 (macroscopic jump cost). The map $u \mapsto u^{1/k}$ is L_k -Lipschitz on $[\beta/2, 1]$ and bounded by 1 below $\beta/2$; on $(0, \beta/2)$, $\frac{1}{1-u} \leq \frac{1}{1-\beta/2} \leq 2$, so $\tilde{\nu}((0, \beta/2)) \leq 2\eta((0, \beta/2)) \leq 2V_\varepsilon(\beta/2)^{-2}$ by Chebyshev. Hence

$$\int_{(0, u_0]} |u^{1/k} - \beta^{1/k}| d\tilde{\nu} \leq \frac{2L_k}{1-\beta} S\sqrt{\varepsilon} + \frac{8V_\varepsilon}{\beta^2}.$$

Step 4 (assembly). Couple the macroscopic Poisson count with W_0 's count by thinning (shared count $\text{Poisson}(\lambda \wedge \lambda_{\text{mac}})$, excess independent; [3], Theorem 10.A), shared jump pairs optimally in the multiplier coordinate against the constant target $\beta^{1/k}$, and leave the small jumps unpaired. Writing $W = e^{a_\varepsilon} P_{\text{mac}} P_{\text{small}}$ and $W_0 = e^{a_0} P_0$ with all products in $[0, 1]$, the telescoping inequality and Wald's identity give

$$\begin{aligned} \mathbb{E} |P_{\text{mac}} P_{\text{small}} - P_0| &\leq \underbrace{\int_{(0, u_0]} |u^{1/k} - \beta^{1/k}| d\tilde{\nu}}_{\text{Step 3}} + \underbrace{|\lambda_{\text{mac}} - \lambda|}_{\text{Step 2}} \\ &\quad + \underbrace{\mathbb{E} |P_{\text{small}} - 1|}_{\text{Step 1}}, \end{aligned}$$

each excess jump on either side changing a product by at most 1. The drift difference obeys the same bound: under $\mathbb{E}[W] = 1$, $a_\varepsilon - a_0 = \int (1 - u^{1/k}) d\tilde{\nu} - \lambda(1 - \beta^{1/k})$ decomposes into the same three pieces. Both multipliers are bounded by e^a (Lemma 2(i)), so for $\varepsilon \leq \varepsilon_0$ with $|a_\varepsilon - a_0| \leq 1$,

$$\begin{aligned} W_1 &\leq \mathbb{E} |W - W_0| \leq e^{a_0+1} \left(\mathbb{E} |P_{\text{mac}} P_{\text{small}} - P_0| + |a_\varepsilon - a_0| \right) \\ &\leq 2e^{a_0+1} [\text{Step 1} + \text{Step 2} + \text{Step 3}]. \end{aligned}$$

Substituting the three displays yields the stated K_∞ up to $O(\varepsilon)$ remainders, which are absorbed for $\varepsilon \leq \varepsilon_0$ as in the statement. $\square$

Remark (Why no hypotheses are needed) A coupling built in log-space would require finite activity and a minimum jump size—a Poisson count must be finite, and a de-tilting factor

must be bounded—and an ε -dependent shell decomposition for the small jumps would surrender a logarithmic factor, $O(\sqrt{\varepsilon} \log(1/\varepsilon))$. In the multiplicative coupling both holes close themselves: infinite activity can only accumulate at $u = 1$, where multiplicative cost vanishes at exactly the rate η measures, and the de-tilting factor appears only on $(0, u_0]$, where it is the fixed constant $2/(1 - \beta)$. The flatness of the exponential map at $-\infty$, fatal to couplings of jump distributions in x -space, is the resource here.

Theorem 12 (Lower bound: the rate $\sqrt{\varepsilon}$ is exact) *Fix (β, C, r) , $k = 1$, and conservation $\mathbb{E}[W] = 1$. For $d \in (0, d_0]$, $d_0 = d_0(\beta)$ small, let $W_{(d)}$ be the compound-Poisson multiplier with tilted measure*

$$\eta_d = \frac{|A|}{2} (\delta_{\beta-d} + \delta_{\beta+d})$$

(jump atoms $\beta \pm d$ with Lévy masses $\frac{|A|/2}{1-(\beta \pm d)}$, drift by conservation). Then:

- (i) *its $A1$ residual satisfies $\frac{|A|}{|\ln r|} d^2 \leq \varepsilon(d) \leq c_2(\beta) \frac{|A|}{|\ln r|} d^2$, where $c_2(\beta) = [e \min_{\rho \in [\beta, (1+\beta)/2]} \rho |\ln \rho|]^{-1} \vee 1$;*
- (ii) *with $\lambda_d = \|\tilde{\nu}_d\| = \lambda/(1 - d^2(1 - \beta)^{-2})$,*

$$W_1(\text{law } W_{(d)}, \text{law } W_0) \geq \lambda_d e^{-\lambda_d} e^{|A|} d \geq c_1(\beta, C, r) \sqrt{\varepsilon(d)}.$$

Hence no propagation bound of order $o(\sqrt{\varepsilon})$ is possible: combined with Theorem 11, the exact rate is $\Theta(\sqrt{\varepsilon})$.

Proof (i) The moments are $\mu_m = \frac{|A|}{2} [(\beta - d)^m + (\beta + d)^m]$, so the signed residuals are

$$\epsilon_m = \frac{\mu_{m+1} - \beta \mu_m}{|\ln r|} = \frac{|A| d}{2 |\ln r|} [(\beta + d)^m - (\beta - d)^m] \geq 0,$$

with $\epsilon_0 = 0$ and $\epsilon_1 = |A|d^2/|\ln r|$ exactly, giving the lower bound on $\varepsilon(d) = \sup_m \epsilon_m$. For the upper bound, the mean value theorem gives $(\beta + d)^m - (\beta - d)^m \leq 2dm(\beta + d)^{m-1}$, and $\sup_{x \geq 0} x \rho^{x-1} = (e\rho |\ln \rho|)^{-1}$ for $\rho = \beta + d \leq (1 + \beta)/2$.

(ii) For $k = 1$ the conservation drift is $a_{(d)} = \int (1 - u) d\tilde{\nu}_d = \|\eta_d\| = |A|$, identical to $a_0 = \lambda(1 - \beta) = |A|$: the family is *drift-rigid*. The support of $\text{law}(W_0)$ is the geometric set $G = \{e^{a_0} \beta^j : j \geq 0\}$. Take the 1-Lipschitz test function

$$f(w) := \min\left(\text{dist}(w, G), \frac{e^{a_0} \beta(1-\beta)}{2}\right) \geq 0,$$

which vanishes on G , so $\mathbb{E}f(W_0) = 0$. On the one-jump event of $W_{(d)}$ (probability $\lambda_d e^{-\lambda_d}$), $W_{(d)} = e^{a_0}(\beta \pm d)$, whose distance to the nearest point of G is exactly $e^{a_0}d$ for $d < \beta(1 - \beta)/2$ (the neighbors e^{a_0} and $e^{a_0}\beta^2$ are farther, and $e^{a_0}d$ is below the cap); all other events contribute ≥ 0 . By Kantorovich–Rubinstein duality [19],

$$W_1 \geq \mathbb{E}f(W_{(d)}) - \mathbb{E}f(W_0) \geq \lambda_d e^{-\lambda_d} e^{a_0} d,$$

and substituting $d \geq (|\ln r| \varepsilon(d) / (c_2(\beta) |A|))^{1/2}$ from (i) gives the $c_1 \sqrt{\varepsilon}$ form. (The same construction works for $k > 1$ on the lattice $\{e^{a_0} \beta^{j/k}\}$; $k = 1$ is stated for cleanliness.) $\square$

Remark (The stability theory is elementary and now complete) The change of variables $u = e^{kx}$ maps the Lévy measure to the compact interval $[0, 1]$, where a second-moment test against the candidate atom decides everything: the classification is the $\varepsilon = 0$ case of the

variance identity in Theorem 10, Step 2; the stability constant is read off two residuals and is sharp; and the propagation to the multiplier law is a telescoped multiplicative coupling with no tail estimates (boundedness, Lemma 2(i)), sharp in rate by Theorem 12. Every quantitative statement in the package—variance constant $(1 + \beta)$, Wasserstein constant, propagation rate $\Theta(\sqrt{\varepsilon})$ —is attained by an explicit family. In particular the log-Poisson class is an open set, with exactly computed modulus, in the space of cascade multiplier distributions metrized by A1 residuals.

6 Beyond independence: stationary and Markov multipliers

The i.i.d. assumption enters the preceding sections through the identity $\mathbb{E}[(W_1 \cdots W_n)^p] = (\mathbb{E}[W^p])^n$, which converts scaling data into one-step moment data. This section determines exactly what survives without it. Let (W_n) be *stationary ergodic*, $X_n = \ln W_n$, $S_n = X_1 + \cdots + X_n$, and define exponents asymptotically:

Standing assumptions (S). For each lattice p : $m_n(p) := \mathbb{E}[e^{pS_n}] < \infty$ for all n , and

$$\Lambda(p) := \lim_{n \rightarrow \infty} \frac{1}{n} \ln m_n(p) \quad \text{exists and is finite;} \quad \zeta_p := \Lambda(p)/\ln r.$$

A1 is imposed on $\delta_p = \zeta_{p+k} - \zeta_p$ as before. Write $\Lambda_{\text{LP}}(p) = ap + \lambda(e^{bp} - 1)$ for the log-Poisson limiting cumulant function with the parameter dictionary of Theorem 3.

Theorem 13 (Asymptotic-statistics classification) *Under (S), A1 holds on the lattice if and only if $\Lambda = \Lambda_{\text{LP}}$ on the lattice. Consequently, under A1 every observable computed from lattice exponents—structure-function exponents and moment ratios—coincides exactly with that of the i.i.d. log-Poisson cascade with parameters (β, γ, C) ; and whenever $\Lambda = \Lambda_{\text{LP}}$ holds on all of $[0, \infty)$ (as in the construction of Theorem 14), the large-deviation rate function and multifractal spectrum coincide as well (see the remark below; lattice A1 alone pins Λ off-lattice only up to convexity).*

Proof Lemma 1 is purely algebraic, so A1 gives $\Lambda(km) = a km + \lambda(e^{b km} - 1)$ for all m ; the converse is the computation of Proposition 5. $\square$

Remark If Λ exists, is finite and differentiable on a neighborhood of $[0, \infty)$, the Gärtner–Ellis theorem [6] yields a large-deviation principle for S_n/n with rate $I(h) = \sup_p [ph - \Lambda(p)]$ on the exposed range—the Legendre structure of Theorem 3(iii). Off-lattice, Λ is pinned between consecutive lattice values by convexity.

Theorem 13 is deliberately easy; the substantive question is whether A1 still determines the multiplier *law*. It does not:

Theorem 14 (Interleaved cascade: the law is not determined) *There exists a stationary ergodic multiplier sequence (W_n) —realizable as a function of a stationary, irreducible, positive-recurrent countable-state Markov chain—such that:*

(i) $m_n(p) = r^{n\zeta_p^{\text{LP}}}$ exactly, where $\zeta_p^{\text{LP}} := \gamma p + C(1 - \beta^{p/k})$ is the log-Poisson exponent curve, for every even n and every real $p \geq 0$; hence (S) holds, $\Lambda = \Lambda_{\text{LP}}$ on all of $[0, \infty)$, and A1 holds exactly, in its strongest (real- p) form;

(ii) the one-step marginal of $\ln W_1$ is not log-Poisson;

(iii) the sequence is not i.i.d., and is not equal in law to any i.i.d. cascade.

Consequently the law-level conclusion of Theorem 3 does not extend beyond independence, by any proof.

Proof Construction. With (a, b, λ) as in Theorem 3 and $\lambda' := 2\lambda$, let $A_1, A_2, \dots$ be i.i.d. $\text{Poisson}(\lambda')$, define

$$Y_{2j-1} = a + bA_j, \quad Y_{2j} = a \quad (j \geq 1),$$

draw a phase $\theta \sim \text{Unif}\{0, 1\}$ independent of everything, and set $X_n := Y_{n+\theta}$, $W_n := e^{X_n}$: jump slots of doubled intensity alternate with deterministic slots.

Stationarity and Markov realization. The pair $Z_n := ((n + \theta) \bmod 2, X_n)$ is a Markov chain on $\{0, 1\} \times (a + b\mathbb{N}_0)$: from phase-0 states the next value is $a + bA$ with fresh $A \sim \text{Poisson}(\lambda')$ and the phase flips; from phase-1 states the next value is a and the phase flips. The chain is irreducible on its reachable set and positive recurrent; the phase-uniform stationary law makes (Z_n) stationary and W_n a function of it.

Ergodicity. Let P_0, P_1 be the path laws given $\theta = 0, 1$, so the law is $\frac{1}{2}(P_0 + P_1)$ with $P_1 = P_0 \circ T^{-1}$ (T = shift). If E is T -invariant, it is T^2 -invariant, and under P_0 the double shift is ergodic (the blocks (Y_{2j-1}, Y_{2j}) are i.i.d.), so $P_0(E) \in \{0, 1\}$; and $P_1(E) = P_0(T^{-1}E) = P_0(E)$. Hence $\mathbb{P}(E) \in \{0, 1\}$.

Exact exponents. For even $n = 2m$ the window $\{1, \dots, 2m\}$ contains exactly m jump slots under either phase, carrying m distinct i.i.d. A_j 's; hence, exactly, for every real p ,

$$\begin{aligned} m_{2m}(p) &= e^{2map} \exp[m\lambda'(e^{bp} - 1)] = \exp[2m(ap + \lambda(e^{bp} - 1))] \\ &= \exp[2m\Lambda_{\text{LP}}(p)]. \end{aligned}$$

For odd n the window covers m or $m+1$ jump slots depending on the phase, and $\frac{1}{n} \ln m_n(p) \rightarrow \Lambda_{\text{LP}}(p)$. Theorem 13 then gives A1 exactly.

Non-log-Poisson marginal. $\mathbb{P}(X_1 = a) = \frac{1}{2}(1 + e^{-2\lambda})$, whereas the $\log\text{-Poisson}(\lambda)$ marginal has $\mathbb{P}(X = a) = e^{-\lambda}$; for She–Lévêque dissipation parameters ($\lambda = 2 \ln 2$): 0.531 versus 0.250. The marginal is the mixture $\frac{1}{2}\delta_a + \frac{1}{2}\text{law}(a + b\text{Poisson}(2\lambda))$. No choice of parameters repairs this: a log-Poisson marginal charges every point of its lattice $a' + b'\mathbb{N}_0$, so support comparison forces $(a', b') = (a, b)$; the atom-mass ratios at depths $j \geq 1$ are those of $\text{Poisson}(2\lambda)$, forcing $\lambda' = 2\lambda$, whose top-atom mass $e^{-2\lambda}$ lies strictly below the mixture's $\frac{1}{2}(1 + e^{-2\lambda})$.

Non-i.i.d. Given $X_n \neq a$ the next multiplier is deterministic: $\mathbb{P}(X_{n+1} = a \mid X_n \neq a) = 1 \neq \mathbb{P}(X_{n+1} = a)$. $\square$

Remark The construction generalizes freely (blocks of length L , arbitrary allocation of the total jump intensity across slots, Markov-modulated loads): A1 is compatible with an infinite-dimensional family of mutually singular stationary ergodic processes, all sharing the log-Poisson asymptotics—exactly as Theorem 13 says they must.

Two rigidity results delimit the boundary of Theorem 14.

Corollary 15 (Exchangeable rigidity) *Let (W_n) be exchangeable with $m_n(p) = r^{n\zeta_p}$ holding exactly for $n \in \{1, 2\}$ and all lattice p , with ζ satisfying A1. Then (W_n) is i.i.d. log-Poisson with parameters (β, γ, C) .*

Proof By de Finetti's theorem [11], (W_n) is conditionally i.i.d. given a random directing measure; let $M_p := \mathbb{E}[W_1^p \mid \text{directing measure}]$. Conditional independence gives $m_1(p) = \mathbb{E}[M_p]$ and $m_2(p) = \mathbb{E}[M_p^2]$, so exactness at $n = 1, 2$ reads $\mathbb{E}[M_p] = r^{\zeta_p}$, $\mathbb{E}[M_p^2] = r^{2\zeta_p}$, whence $\text{Var}(M_p) = 0$: $M_p = r^{\zeta_p}$ a.s., for every lattice p simultaneously. Almost every directing measure therefore has exactly the A1 lattice moments, hence equals the log-Poisson law by Lemma 2 and Theorem 3(ii); the mixture is degenerate. $\square$

Theorem 16 (Finite-state impossibility) *Let ξ be an irreducible finite-state Markov chain, stationary, and $W_n = e^{f(\xi_n)}$ with f non-constant. Then the cascade cannot satisfy A1 in its real- p form—the closed form $(**)$ valid for all real $p \geq 0$ —with nontrivial intermittency: there are no parameters (β, γ, C) with $C > 0$ and $\zeta_p = \gamma p + C(1 - \beta^{p/k})$ for all real $p \geq 0$.*

Proof For finite irreducible chains, $\Lambda(p) = \ln \rho(M(p))$ with $M(p)_{xy} = P_{xy} e^{pf(y)}$ and ρ the Perron root—a simple eigenvalue for every real p , hence real-analytic on $\mathbb{R}$. If the closed form held on $[0, \infty)$, then $\Lambda(p) = ap + \lambda(e^{bp} - 1)$ there with $\lambda = -C \ln r > 0$; both sides are real-analytic on $\mathbb{R}$ and agree on an interval, hence agree everywhere (identity theorem). Now let $p \rightarrow -\infty$. Since $f_{\min} = \min_s f(s) > -\infty$, every row sum of $M(p)$ is at most $e^{pf_{\min}}$ for $p \leq 0$, so $\Lambda(p) \leq pf_{\min}$: a linear upper bound. But with $b < 0$,

$$\Lambda_{\text{LP}}(p) - pf_{\min} \geq \lambda e^{b|p|} - \lambda + p(a - f_{\min}) \rightarrow +\infty :$$

contradiction. $\square$

Remark (The boundary, and what A1 really is) The mechanism of Theorem 16 is that finite alphabets make $\ln W$ bounded below, while the log-Poisson generator is intrinsically unbounded below ($\ln W \in a + b\mathbb{N}_0$): A1 forces multipliers with arbitrarily severe attenuation events, and the counterexample of Theorem 14 necessarily has unbounded-below $\ln W$. Assembled, this section says: A1 is an *asymptotic-statistics* axiom. It pins the limiting cumulant function, the spectrum, and the large deviations to the log-Poisson cascade (Theorem 13); it cannot pin the per-step law (Theorem 14); and the i.i.d. log-Poisson cascade is the canonical realization—unique under exchangeability (Corollary 15), with finite-state Markov realizations impossible (Theorem 16). Whether Theorem 16 persists under lattice-only A1 remains open (the identity-theorem step is unavailable on a discrete set; see Section 9).

7 Continuous cascades: log-infinitely-divisible multifractal measures

We now place the theory in the continuous category of Bacry–Muzy multifractal random measures [2, 15], which contains the log-normal multifractal random walk, log-stable measures, and the Barral–Mandelbrot compound Poisson cascades [4] as special cases. We use two structural properties of the class as axioms:

(M1) Exact stochastic scale invariance. For every $\sigma \in (0, 1)$ and $t \leq T$ (the integral scale),

$$(M(\sigma t))_t \stackrel{d}{=} \sigma e^{\Omega_\sigma} (M(t))_t,$$

with Ω_σ infinitely divisible, independent of M , and $\mathbb{E}[e^{q\Omega_\sigma}] = \sigma^{-\psi(q)}$, where ψ is the Lévy exponent of the generator per unit logarithmic scale.

(M2) Conservation. $\mathbb{E}[e^{\Omega_\sigma}] = 1$, i.e. $\psi(1) = 0$.

(S_c) $\psi(q) < \infty$ for all $q \geq 0$ (the continuous analogue of finite multiplier moments).

From (M1), wherever $\mathbb{E}[M([0, t])^q] < \infty$,

$$\mathbb{E}[M([0, t])^q] \propto t^{\zeta_q}, \quad \zeta_q = q - \psi(q),$$

and moments of the total mass are finite (for $q > 1$) precisely on the window where $\zeta_q > 1$ [2]. Which orders are usable thus depends on the generator: when ζ_q eventually decreases (log-normal, log-stable) only a finite window $[0, q^*)$ of finite moments exists, whereas the She–Lévêque curve itself is strictly increasing under (M2) whenever $\gamma \geq 0$ (at She–Lévêque values $\gamma = 1/3$), so every moment of the compound Poisson cascade is finite. The operative limit on structure-function data is statistical: high-order sample moments are noise-dominated, and in practice exponents are trusted only on a short initial window, typically $q \lesssim 3$. The hierarchical symmetry A1 is imposed on $\delta_q = \zeta_{q+k} - \zeta_q$ as before.

The dictionary to the discrete theory is one line: with $\phi_c(q) := \psi(q+k) - \psi(q)$ one has $\delta_q = k - \phi_c(q)$, and A1 gives, by Lemma 1,

$$\phi_c(mk) - \phi_{c,\infty} = A_c \beta^m, \quad A_c = -(\delta_0 - \delta_\infty) = -C(1 - \beta) < 0 :$$

formally the discrete identities with $\ln r \mapsto -1$, so $|A_c| = C(1 - \beta)$ and the rate $\lambda = -C \ln r$ becomes the intensity C per unit log-scale. Every $|\ln r|$ in the discrete constants disappears.

Theorem 17 (Four orders do not identify the class; five contiguous orders do) *Fix the She–Lévêque exponent curve $\zeta_q^{\text{SL}} = \gamma q + C(1 - \beta^{q/k})$, $\beta \in (0, 1)$, $C > 0$, normalized by (M2).*

- (i) (Impersonation up to four orders.) *Assume additionally $\gamma \geq 0$, as at She–Lévêque values, so that the matched curve keeps the measures nondegenerate (every impersonator below matches $\zeta_{2k} = \zeta_{2k}^{\text{SL}} > 1$). For every lattice window $F \subseteq \{0, k, 2k, 3k\}$ there is a continuum of exactly scale-invariant log-ID multifractal random measures whose generators are not compound Poisson with a single atom—in particular are not the compound Poisson cascade—yet whose exponents satisfy $\zeta_q = \zeta_q^{\text{SL}}$ for every $q \in F$. Moreover every impersonator of the full window $\{0, k, 2k, 3k\}$ is strictly more intermittent at the next order: $\zeta_{4k} < \zeta_{4k}^{\text{SL}}$, with $\zeta_{4k}^{\text{SL}} - \zeta_{4k} \leq C\beta(1 - \beta)^3$ for generators without amplifying jumps.*
- (ii) (Five-point rigidity.) *Let M satisfy (M1), (M2) with generator triplet $(\tilde{a}, \sigma_0^2, \Pi)$ and Lévy exponent ψ finite on $[0, 4k]$ (weaker than (S_c)). If $\zeta_q = \zeta_q^{\text{SL}}$ at the five contiguous points $q \in \{0, k, 2k, 3k, 4k\}$ —in particular on any window $F \supseteq$*

$\{0, k, 2k, 3k, 4k\}$ —then $\sigma_0^2 = 0$ and $\Pi = C\delta_b$ with $b = (\ln \beta)/k$: M is the compound Poisson cascade. No impersonator exists.

Structure-function data on the practically accessible window (typically $q \lesssim 3$: part (i)) cannot certify the log-Poisson class; five exact contiguous exponent values (part (ii)) can.

Proof Throughout take $k = 1$ (general k is identical after $u = e^{kx}$) and work in the tilted coordinates $u = e^x$, in which the CPC generator (Theorem 18) has $\tilde{\Pi} = C\delta_\beta$. The key object is the *squared-tilt* measure

$$\rho := (u - 1)^2 d\tilde{\Pi} \geq 0 \quad \text{on } (0, \infty) \setminus \{1\}, \quad r_q := \int u^q d\rho;$$

because $(u - 1)^2$ is a square, ρ is nonnegative on *both* sides of $u = 1$. Finiteness of r_0, r_1, r_2 follows from $\psi(4) < \infty$: near $x = 0$ the integrand is $\asymp x^2 u^q$ (Lévy integrability); on $x \leq -1$ it is bounded; on $x \geq 1$ it is $\leq e^{(q+2)x}$ with $q + 2 \leq 4$. Taking the step-one second difference of the Lévy–Khinchine formula, the drift and compensator terms cancel exactly, the Gaussian term contributes σ_0^2 , and $e^{(m+2)x} - 2e^{(m+1)x} + e^{mx} = e^{mx}(e^x - 1)^2$, so

$$\Delta^2 \psi(m) = \psi(m+2) - 2\psi(m+1) + \psi(m) = \sigma_0^2 + r_m, \quad (3)$$

while on the She–Lévêque side $\Delta^2 \psi^{\text{SL}}(m) = C\beta^m(1 - \beta)^2 =: A'\beta^m$.

(ii), *five-point rigidity*. Matching ζ (hence ψ) at the five contiguous points $\{0, 1, 2, 3, 4\}$ matches the three second differences $m = 0, 1, 2$ of (3): $\sigma_0^2 + r_m = A'\beta^m$. Test against the square $(u - \beta)^2$:

$$\begin{aligned} 0 &\leq \int (u - \beta)^2 d\rho = r_2 - 2\beta r_1 + \beta^2 r_0 \\ &= A'(\beta^2 - 2\beta^2 + \beta^2) - \sigma_0^2(1 - 2\beta + \beta^2) \\ &= -\sigma_0^2(1 - \beta)^2 \leq 0. \end{aligned}$$

Hence $\sigma_0^2 = 0$ and $\int (u - \beta)^2 d\rho = 0$, so ρ vanishes off $\{\beta\}$ and $\rho = r_0\delta_\beta = A'\delta_\beta$. Inverting the tilt $((u - 1)^2 > 0$ off $u = 1$, and $\tilde{\Pi}(\{1\}) = \Pi(\{0\}) = 0$ by the Lévy convention) gives $\tilde{\Pi} = A'(1 - \beta)^{-2}\delta_\beta = C\delta_\beta$, i.e. $\Pi = C\delta_b$; (M2) pins $\tilde{a}$ and $\psi \equiv \psi^{\text{SL}}$. The identification with the Barral–Mandelbrot cascade is [2]. Note no divergence argument and no moments beyond $q = 4$ are used: the squared tilt is sign-definite across $u = 1$, so amplifying jumps ($u > 1$) and the Gaussian component die on the finite window.

(i), *construction at four orders*. For the physically typical window $F = \{0, 1, 2\}$ perturb the CPC:

$$\tilde{\Pi}_s = (C - sc)\delta_\beta + s(w_1\delta_{v_1} + w_2\delta_{v_2}), \quad 0 < v_1 < \beta < v_2 < 1,$$

with the drift re-fixed by (M2) for each s , and impose $\sum_i w_i(v_i^m - 1) = c(\beta^m - 1)$ for $m = 1, 2$: the $m = 1$ equation makes the (M2)-drifts coincide, the $m = 2$ equation matches ζ_2 , and $\zeta_0 = 0$, $\zeta_1 = 1$ are automatic. At $(\beta, v_1, v_2) = (2/3, 0.3, 0.9)$, $c = 1$, the 2×2 system gives $w_1 = 5/27 \approx 0.1852$, $w_2 = 55/27 \approx 2.0370$, both *positive*, so $\tilde{\Pi}_s \geq 0$ for all $s \in [0, C/c]$: a one-parameter family of genuine Lévy measures, none a single atom for $s > 0$, all matching ζ^{SL} exactly on F (and differing beyond: at $m = 3$ the perturbed exponent differs by $+77/5400 \approx +0.0143$ at $s = \frac{1}{2}$).

This two-sided ansatz, however, does *not* extend past $F = \{0, 1, 2\}$: if $\mu^+ = \sum_i w_i\delta_{v_i} \geq 0$ with $v_i \in (0, 1)$ satisfies the constraints for $m = 1, 2, 3$, then with $d\sigma := (1 - u)d\mu^+ \geq 0$ and $1 - u^m = (1 - u)(1 + u + \dots + u^{m-1})$ the constraints give $s_j := \int u^j d\sigma = c(1 - \beta)\beta^j$

for $j = 0, 1, 2$, whence $\int (u - \beta)^2 d\sigma = 0$ and $\mu^+ = c\delta_\beta$: the perturbation collapses to the CPC atom. The maximal window $\{0, 1, 2, 3\}$ is reached instead by a one-sided construction in ρ -space: solve $r_0 = A'$, $r_1 = A'\beta$ with two atoms, e.g. $\rho = w_1\delta_{0.4} + w_2\delta_{0.95}$, $(w_1, w_2) = (0.114478, 0.107744)$ at $(\beta, C) = (2/3, 2)$, i.e. $\tilde{\Pi} = \rho/(1-u)^2 = 0.31799\delta_{0.4} + 43.0976\delta_{0.95}$ with (M2)-drift $\tilde{a} = 2.345679$. By (3) this matches ζ^{SL} at $q = 0, 1, 2, 3$ exactly (with $\zeta_2 = \frac{16}{9} > 1$: a nondegenerate multifractal measure), is not single-atom compound Poisson, and deviates at $q = 4$ by $-(r_2 - A'\beta^2) = -0.01679$. Varying the atom positions $0 < v_1 < \beta < v_2 < 1$ solves the same two constraints with strictly positive weights $w_1 = A'(v_2 - \beta)/(v_2 - v_1)$, $w_2 = A'(\beta - v_1)/(v_2 - v_1)$, giving a two-parameter continuum of impersonators of the full window.

(i), *one-sidedness at the next order*. For any generator matching ζ^{SL} on $\{0, 1, 2, 3\}$, (3) at $m = 0, 1$ gives $\sigma_0^2 + r_0 = A'$ and $\sigma_0^2 + r_1 = A'\beta$, and

$$\zeta_4 - \zeta_4^{\text{SL}} = -(\sigma_0^2 + r_2 - A'\beta^2).$$

Cauchy–Schwarz $r_1^2 \leq r_0 r_2$ forces $\sigma_0^2 + r_2 \geq A'\beta^2$, with equality iff $\sigma_0^2 = 0$ and $\rho = A'\delta_\beta$ —i.e. iff the generator is the CPC itself. Hence every genuine impersonator strictly undershoots at $q = 4$; for attenuation-only generators (no atoms $u > 1$) the deficit is at most $A'\beta(1 - \beta) = C\beta(1 - \beta)^3$ ($= 4/81 \approx 0.049$ at She–Lévêque values), while amplifying atoms make it unbounded as $u_{\max} \rightarrow \infty$. $\square$

Remark The divergence steps of Theorem 7 need $q \rightarrow \infty$, and on a three-point window even $\sigma_0^2 > 0$ survives (a quadratic log-normal exponent interpolates any three-point window with the correct convexity)—but the obstruction stops there. The squared tilt $\rho = (u - 1)^2 d\tilde{\Pi}$ of part (ii) is sign-definite across $u = 1$, so the Gaussian component and amplifying jumps die on a finite window with no divergence bookkeeping: identifiability does *not* require constraints of unbounded order—five exact contiguous orders suffice. (Version 2 of this paper asserted the contrary; part (ii) corrects it.) What structure functions cannot supply in practice is exactness at five orders inside the statistically accessible window, typically $q \lesssim 3$: the barrier to certification by structure functions is statistical, not one of principle. The scale-invariance factor Ω_σ remains the robust observable: under (Sc) it has all exponential moments, at every σ , and its statistics (“magnitude” statistics, in the language of [5]) are observable at all orders.

Theorem 18 (Generator-level classification: A1 selects the compound Poisson cascade) *Let M satisfy (M1), (M2), (Sc), with nontrivial intermittency $C > 0$, and define the generator exponents $\zeta_q = q - \psi(q)$ for all $q \geq 0$. Then A1 on the full lattice $k\mathbb{N}_0$ holds if and only if*

$$\sigma_0^2 = 0, \quad \Pi = C\delta_b, \quad b = \frac{\ln \beta}{k} < 0,$$

with drift $\tilde{a} = C(1 - \beta^{1/k})$ fixed by (M2); equivalently

$$\Omega_\sigma \stackrel{d}{=} \tilde{a} \ln(1/\sigma) + b \text{Poisson}(C \ln(1/\sigma)) \quad \text{for every } \sigma \in (0, 1),$$

log-Poisson at every scale ratio simultaneously, and M is the Barral–Mandelbrot compound Poisson cascade [4] with fixed multiplier atom $e^b = \beta^{1/k}$ and Poisson intensity C on the time-log-scale cone. No other member of the Bacry–Muzy class—log-normal, log-stable, or any intermediate—satisfies A1.

Proof Reverse: with $\Pi = C\delta_b$, $\psi(q) = \tilde{a}q + C(e^{bq} - 1)$, so $\zeta_q = q - \psi(q) = (1 - \tilde{a})q + C(1 - \beta^{q/k})$, which satisfies A1 by Proposition 5 with $\gamma = 1 - \tilde{a}$; (M2) gives $\tilde{a} = C(1 - \beta^{1/k})$, the continuous conservation constraint ($\zeta_1 = 1$).

Forward: ψ is a Lévy–Khintchine exponent, finite for all $q \geq 0$, and $\phi_c(q) = \psi(q+k) - \psi(q)$ has a finite limit under A1: exactly the hypothesis configuration of Theorem 7’s proof under the dictionary $\ln r \mapsto -1$. Steps 2, 3, $3\frac{1}{2}$ and 4 of that proof apply verbatim: a Gaussian component, positive jumps, or $\int_{|x| \leq 1} |x| d\Pi = \infty$ each force $\delta_q = k - \phi_c(q) \rightarrow -\infty$; then with $\eta = (1 - u) d\tilde{\Pi}$, $u = e^{kx}$, A1 gives $\int u^m d\eta = |A_c| \beta^m$ and the variance identity yields $\eta = |A_c| \delta_\beta$, i.e. $\tilde{\Pi} = \frac{|A_c|}{1 - \beta} \delta_\beta = C\delta_\beta$ and $\Pi = C\delta_b$. The identification of the compound-Poisson member of the Bacry–Muzy class with the Barral–Mandelbrot cascade is [2].

Alternatively, and with weaker hypotheses, the forward direction is a corollary of Theorem 17(ii): A1 on the full lattice gives the closed form $\zeta_q = \zeta_q^{\text{SL}}$ on $\{0, k, 2k, 3k, 4k\}$ (Lemma 1 under the dictionary), and five-point rigidity forces the CPC—using only ψ finite on $[0, 4k]$ rather than (S_c) , and none of the divergence steps. $\square$

Corollary 19 (Stability with native constants) *In the setting of Theorem 18, if A1 holds to within ε on the generator lattice—with d^* the true limit (reading (i)) or fitted (reading (ii))—then $\sigma_0^2 = 0$, Π is supported on $(-\infty, 0]$, and with $\eta = (1 - u) d\tilde{\Pi}$, $\|\eta\| = |A_c| = C(1 - \beta)$ exactly:*

$$W_1\left(\frac{\eta}{\|\eta\|}, \delta_\beta\right) \leq \sqrt{\frac{1 + \beta}{C(1 - \beta)}} \sqrt{\varepsilon} \quad (\text{i}), \quad \leq \sqrt{\frac{2}{C(1 - \beta)}} \sqrt{\varepsilon} \quad (\text{ii}),$$

both sharp; for She–Lévéque values $(\beta, C) = (2/3, 2)$: $1.58\sqrt{\varepsilon}$ and $1.73\sqrt{\varepsilon}$. The propagation to the law of the per-octave factor $e^{\Omega_{1/2}}$ holds at the unconditional sharp rate $\Theta(\sqrt{\varepsilon})$ of Theorems 11–12 (whose proofs nowhere used finite activity—the natural situation here).

Proof Theorem 10, Steps 0–3, under the dictionary $\ln r \mapsto -1$, $|A| \mapsto |A_c| = C(1 - \beta)$; the sharpness families transfer symbol-for-symbol, as does the propagation argument of Section 5. $\square$

Remark (What the two theorems mean together) Theorems 17 and 18 are two halves of one statement about observability: on the practically accessible window (typically $q \lesssim 3$) structure functions cannot identify the cascade class, while five exact contiguous orders—or, at the generator level, magnitude statistics at three lattice orders—classify it completely, with quantitative stability. This places a theorem under the long-standing practical preference for magnitude-cumulant analysis over high-order structure functions [5], now as a statement about statistics rather than principle: the structure-function route fails not because finite-order data can never certify the class, but because exactness at the five orders it would need is beyond finite-sample resolution. Note also that the continuous category is *more* rigid than the discrete one: infinite divisibility, an assumption in Theorem 7, is automatic here (consistency of $\Omega_{\sigma\sigma'} \stackrel{d}{=} \Omega_\sigma + \Omega'_{\sigma'}$), so the classification needs no distributional hypothesis beyond membership in the class—and by Theorem 17(ii) it needs only five lattice moments, with no divergence bookkeeping. Finally, the most-singular-branch geometry (Corollary 4) reads natively: the probability that the cone above a point carries no Poisson point down to scale ℓ is ℓ^C —codimension C , with no discretization anywhere.

8 Corollaries

Corollary 20 (Conservation constraint) *If there exists an index $k_0 > 0$ such that $\zeta_{k_0} = z_0$ for a known constant z_0 fixed by an exact conservation law, then*

$$\gamma = \frac{z_0 - C(1 - \beta^{k_0/k})}{k_0}.$$

This reduces the observable parameters from two to one.

Corollary 21 (Codimension identification) *If the most singular structures have Hausdorff codimension C_{geom} and $C = C_{\text{geom}}$, then, together with the conservation constraint of Corollary 20, β alone determines the full exponent curve, the multifractal spectrum, and the cascade distribution.*

Corollary 22 (Spectrum width) *The width of the multifractal spectrum is*

$$\Delta h = h_{\max} - h_{\min} = \frac{C}{k} |\ln \beta|,$$

where $h_{\max} = \gamma + (C/k)|\ln \beta|$ (at $p = 0$, most regular) and $h_{\min} = \gamma$ (at $p \rightarrow \infty$, most singular).

Corollary 23 (Parameter-free stability constant) *Since $|A| = C(1 - \beta)|\ln r|$, the scale ratio cancels in Theorem 10:*

$$W_1\left(\frac{\eta}{\|\eta\|}, \delta_\beta\right) \leq \sqrt{\frac{1 + \beta}{C(1 - \beta)}} \sqrt{\varepsilon} \quad (\text{i}), \quad \leq \sqrt{\frac{2}{C(1 - \beta)}} \sqrt{\varepsilon} \quad (\text{ii}),$$

independent of r and k (and identical to the native continuous constants of Corollary 19). For fully developed turbulence ($\beta = 2/3$, $C = 2$, in either the dissipation form $k = 1$ or the velocity form $k = 3$) the constants are $\sqrt{5/2} \approx 1.58$ and $\sqrt{3} \approx 1.73$: a measured violation ε of hierarchical symmetry confines the normalized tilted jump measure within $1.58\sqrt{\varepsilon}$ (resp. $1.73\sqrt{\varepsilon}$) of $\delta_{2/3}$ in Wasserstein-1 distance.

9 Concluding remarks

The results of this paper show that the hierarchical symmetry A1 carries considerably more force than might be expected from its appearance as a simple linear recurrence. Within i.i.d. multiplicative cascades it is equivalent to the log-Poisson class, with sharp stability constants and the exact propagation rate $\Theta(\sqrt{\varepsilon})$; beyond independence it pins all asymptotic statistics (and provably nothing more); and in the continuous category it selects exactly the compound Poisson cascade at the generator level, while structure functions can do so only from five exact contiguous moment orders—beyond the window accessible in practice. The following directions remain open.

1. **Lattice-only finite-state rigidity.** Theorem 16 assumes the closed exponent form for all real $p \geq 0$; under lattice-only A1 the identity-theorem step is unavailable, and Carlson-type interpolation is blocked by possible complex eigenvalue crossings of the tilted transfer matrix. We expect the conclusion to persist.

2. **Determination of k , and the joint-in- k test.** The hierarchy step k is treated as given; in applications it must be estimated. A1 at several steps simultaneously imposes the compatibility constraint $\ln \beta(k) \propto k$, and the statistical gain from the joint test is unquantified.
3. **Boundary cases.** The log-normal class is the $\beta \rightarrow 1$ closure point of the log-Poisson family ($b \rightarrow 0$, $\lambda b^2 \rightarrow \sigma^2$): quantifying the degeneration of identifiability as $\beta \uparrow 1$ would unify the classification with its principal rival. The maximal-intermittency limit $\beta \rightarrow 0$ likewise deserves analysis.
4. **Statistics of the A1 test.** The present results are exact-population statements. A finite-sample theory—error bars on $(\hat{\beta}, \hat{d}^*)$, power against log-normal and log-stable alternatives, with the reading-(ii) constants of Theorem 10 and Corollary 19 as the operative null band—is the missing link between the classification and data; the window obstruction of Theorem 17(i) dictates that such a theory be built on magnitude statistics rather than high-order structure functions. (By Theorem 17(ii) this is a dictate of statistics, not principle: five exact contiguous orders would suffice, but exactness at five orders is precisely what finite noisy data cannot supply.) Sparse lattice windows—neither contained in $\{0, k, 2k, 3k\}$ nor containing $\{0, k, 2k, 3k, 4k\}$ —are not settled by Theorem 17 and remain open.

Appendix A. Formalization status

The mathematical core of this paper is formalized and kernel-checked in Lean 4 with Mathlib (repository directory `formalization/`, sixteen modules): **zero sorry, zero custom axioms** — every compiled constant depends only on Lean’s three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), enforced by a whole-library environment walk (`AxiomAudit.lean`) that fails on any violation and thereby also soundly excludes every escape hatch (`sorry` introduces the axiom `sorryAx`). Toolchain and Mathlib revision are pinned; `./check.sh` reproduces the full gate, and `Freeburg/Main.lean` is a compile-checked manifest of the principal claims (it fails to build if any is renamed or removed).

Machine-checked (Lean names in the repository’s mapping tables): Lemma 1, with the lattice-level skeleton of Corollary 6 (`satisfiesA1_iff_closed_form`); Proposition 5 (`logPoisson_satisfiesA1`); Lemma 2 (`lemma2_boundedness_ext_of_lattice_moments_eq`); Theorem 3(ii) at the level of laws (`theorem3_law_uniqueness_W`); Theorem 7, forward (`theorem7_forward_nu`); Theorem 10 under both readings with both sharp constants attained exactly (`theorem10_variance_known_fitted`, `variance_le_sharp`, `fitted_variance_le_sharp`, Wasserstein form `theorem10_wasserstein_W1`); Theorem 11, unconditional (`theorem11_rate_upper`); Theorem 12 (`theorem12_rate_lower`); the $\Theta(\sqrt{\varepsilon})$ bracket (`theorem11.12_theta`); and, in this version, Theorem 17 in both parts — five-point rigidity (`theorem17_five_point_rigidity`), one-sidedness with its equality case and deficit budget (`theorem17_one_sided`, `theorem17_one_sided_budget`), and the exact-rational impersonators including the injective three-atom continuum (`theorem17_impersonation_three_point`, `nu3_inj0n`),

`theorem17_impersonation_four_point`) — together with Theorem 18’s forward direction by the five-moment route under hypotheses weaker than (S_c) (`theorem18_forward_fiveMoment`). Section 6 is checked in scoped form: Theorem 13 in full (`theorem13_asymptotic_classification`); Theorem 14’s quantitative core—exact even-block log-Poisson moments and the strict marginal mismatch $\frac{1}{2}(1 + e^{-2\lambda}) > e^{-\lambda}$ (`theorem14_even_block_moments`, `theorem14_marginal_ne_logPoisson`); Corollary 15 in conditional form—the de Finetti directing kernel as hypothesis, after which variance-vanishing and point-wise determinacy are fully checked (`corollary15_conditional`); and Theorem 16’s growth-contradiction core (`theorem16_scoped`). Corollaries 19–23 are checked (`corollary20_conservation` through `corollary23_native_stability`, the latter carrying Corollaries 19 and 23 as a single statement, with the spectrum width of Corollary 22 obtained from a kernel-checked derivative).

Strengthenings found by formalization (each documented in the repository): the converse needs no $\lambda > 0$; the Dirac rigidity is uniform in β and needs no sign hypothesis on the intensity ($C \geq 0$ is a conclusion); the variance-level sharp constants are attained exactly on the closed interval, not only in the limit; Theorem 12(ii) holds for every target rate and drift; Theorem 11’s constant improves at $k = 1$ and never consumes conservation; the one-sided deficit budget needs no $\sigma^2 \geq 0$. The formalization also caught one sign typo (Theorem 12’s parenthetical, fixed before submission of v2).

Still informal, and why: the classical inputs that the formalized statements deliberately take as hypotheses rather than re-prove, pending Mathlib infrastructure: the Lévy–Khinchine representation theorem (the log-ID class is formalized *by triplet*, exactly as Theorems 7, 10, 17 and 18 state their hypotheses); de Finetti’s theorem (Corollary 15’s directing kernel); Perron–Frobenius simplicity with analytic perturbation and the identity theorem (Theorem 16’s extension of Λ to $\mathbb{R}$); the Bacry–Muzy construction of the exactly scale-invariant measure for a given generator (§7 results are stated at the generator level, as their printed proofs argue); and Theorem 14’s stationary-ergodic sequence-space realization together with the Gärtner–Ellis remark (its block-level moments and marginal mismatch are checked). Minor items riding on these bridges (Theorem 3(iii), Corollaries 4 and 8, Proposition 9(b)) and the sparse-window question remain informal. One definitional dictionary is likewise classical: the formalization states W_1 in Kantorovich–Rubinstein dual form, whose equality with transport W_1 on compact supports is classical duality (lower bounds transfer unconditionally). The boundary between machine-checked and informal is exactly as stated here; no other unformalized input is used.

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